
Session 3 — Linear Algebra I

What today is about

Linear algebra is the mathematics of *data in many dimensions* and of *linear transformations*. It is the language of almost every model you will meet later.

- A *vector*: a point, an arrow, or a list of features.
- The *dot product*: similarity, length, angle, and projection.
- A *matrix*, read two ways: a data table *and* a transformation.
- *Linear systems*, the *inverse*, and *rank* — the true dimensionality of data.

Everything is concrete: we will draw almost every idea as a picture.

The single operation everything is built from

Linear combination

Take some vectors, scale each by a number, and add them up:

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k.$$

Almost every concept today — span, matrix multiplication, and next session's PCA — is really a statement about linear combinations.

If this operation feels natural by the end of the session, everything else follows.

A vector, read three ways

Definition

A **vector** $v \in \mathbb{R}^n$ is an ordered list of numbers. We can read it as a *point* at those coordinates, an *arrow* from the origin to that point, or a *list of features* describing one object.

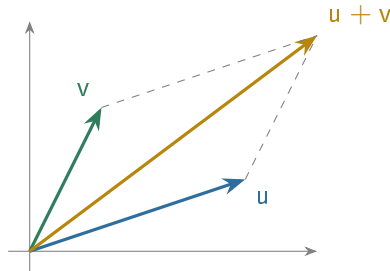
All three readings matter. “Arrow” gives us geometry (length, direction, angle). “Point” lets us picture a whole dataset as a cloud of points. “List of features” is how real data arrives: a participant is a vector of (age, reaction time, accuracy, ...); an image is a vector of pixel values.

Adding and scaling vectors

We combine vectors in just two ways:

- **Add** them componentwise — geometrically, place them tip-to-tail.
- **Scale** them by a number — stretch the arrow, or flip it if the number is negative.

Every richer operation is built from these two.



The dot product: one operation, three meanings

Definition

The **dot product** multiplies matching components and adds:

$$u \cdot v = \sum_i u_i v_i = \|u\| \|v\| \cos \theta,$$

where θ is the angle between the arrows.

From it we also read off *length* $\|v\| = \sqrt{v \cdot v}$ and *distance* $\|u - v\|$. So a single number quietly encodes similarity, size, and angle all at once.

The dot product is a similarity score

Intuition

Vectors pointing the same way have a large positive dot product; perpendicular vectors have dot product zero (“nothing in common”); opposite vectors have a large negative one.

Divide the dot product by the two lengths and you get $\cos \theta$ — the **cosine similarity** used everywhere to compare feature vectors, because it ignores their overall magnitude and keeps only their direction.

Dot product — worked example

Example

Let $u = (1, 2)$ and $v = (3, 1)$.

The dot product is $u \cdot v = 1 \cdot 3 + 2 \cdot 1 = 5$. The lengths are $\|u\| = \sqrt{5}$ and $\|v\| = \sqrt{10}$. So

$$\cos \theta = \frac{5}{\sqrt{5} \sqrt{10}} = \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \theta = 45^\circ.$$

The two arrows sit at a 45° angle — a moderate positive similarity.

A few facts about the dot product

Useful fact

Cauchy–Schwarz: $|u \cdot v| \leq \|u\| \|v\|$ — which is why cosine similarity always lies in $[-1, 1]$.

Triangle inequality: $\|u + v\| \leq \|u\| + \|v\|$ — a detour is never shorter than the direct route.

Unit vector: $\hat{u} = u / \|u\|$ keeps only the direction.

Orthogonality: the geometry of “unrelated”

Key idea

Two vectors are *perpendicular (orthogonal)* exactly when their dot product is zero. This is the geometric meaning of “no linear relationship” — and it is the same idea as zero correlation once data are mean-centred.

It connects directly to last session’s regression: the best-fit line is the one whose residual vector is orthogonal to the predictors. That is the bridge we make concrete next.

Projection: the geometry of least squares

Useful fact

The projection of b onto the direction of a is

$$\text{proj}_a b = \frac{a \cdot b}{a \cdot a} a,$$

and the leftover (the *residual*) is orthogonal to a .

The projection is the “shadow” of b along the direction of a ; the residual is the part of b that a simply cannot explain. Fitting a regression line is exactly projecting the data onto the space of predictors.

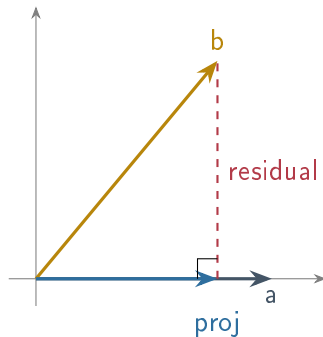
Projection — picture and example

Example

Project $b = (2, 3)$ onto $a = (1, 1)$:

$$\frac{a \cdot b}{a \cdot a} a = \frac{5}{2}(1, 1) = (2.5, 2.5).$$

The residual $(-0.5, 0.5)$ satisfies $a \cdot (-0.5, 0.5) = 0$ — perpendicular, as it must be.



Link to cognitive science

Representing stimuli, words, or neural population states as vectors — and comparing them by dot product or cosine similarity — is the backbone of *representational similarity analysis* in neuroscience and of *word embeddings* in computational linguistics.

“King — man + woman $\approx$ queen” is literally vector arithmetic. A population of neurons firing is a single point in a high-dimensional space, and how two conditions differ is just a distance between two such points.

Practice — vectors and the dot product

Practice — try it

Let $u = (3, 4)$ and $v = (4, -3)$.

1. Find $\|u\|$.
2. Compute $u \cdot v$. Are the two vectors orthogonal?
3. Give the unit vector in the direction of u .

A matrix has two readings — keep both

Definition

An $m \times n$ **matrix** is a rectangular grid of numbers. Read it as a *data table* (rows are observations, columns are features) *and* as a *linear transformation* $x \mapsto Ax$ that moves every vector to a new place.

The data-table reading is how we *store* observations. The transformation reading is how models *act* on them — for example, a layer of a neural network multiplies its input by a weight matrix.

A matrix transforms space

Intuition

A matrix sends the whole grid of vectors to a new grid, keeping the origin fixed and straight lines straight. The *columns* tell you exactly where the basis vectors land: the first column is where $(1, 0)$ goes, the second where $(0, 1)$ goes. Everything else follows by linear combination.

Rotations, reflections, scalings, and shears are all just different matrices — different instructions for where the two axis arrows should end up.

Matrix times vector = combination of columns

The product Ax is a linear combination of the columns of A , weighted by the entries of x :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} a \\ c \end{pmatrix} + y \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

A rotation

$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ sends $(1, 0)$ to $(0, 1)$: the arrow pointing right becomes the arrow pointing up — a quarter turn. The columns of R are where the two axis vectors land.

Multiplying two matrices = composing transformations

The product AB means “do B , then A ”. Entry (i,j) is the dot product of row i of A with column j of B .

Example

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}.$$

Right-multiplying by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ swapped the *columns* of the first matrix.

The algebra of matrices — and one warning

Useful fact

Matrix multiplication is *associative* and *distributive*; $(AB)^T = B^T A^T$; and $AI = A$.

Common pitfall

Matrix multiplication is **not commutative**: in general $AB \neq BA$. Order encodes the order of operations — “rotate then stretch” differs from “stretch then rotate”. Also, inner dimensions must match: an $m \times n$ times an $n \times p$ gives $m \times p$.

Practice — try it

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ (a horizontal shear) and $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

1. Compute Ax .
2. Compute AA (apply the shear twice).
3. Does A change area? Find $\det A$ (we build this next).

A system of equations is one matrix equation

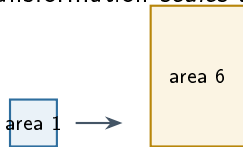
Definition

The system $\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$ is written compactly as $Ax = b$. Solving it means finding the vector x that A maps to b .

Two ways to picture it. *Row picture*: each equation is a line, and the solution is where the lines cross. *Column picture*: which combination of A 's columns produces b ?

A gentle detour: how much does a matrix stretch area?

Before we can “undo” a matrix, one number tells us a lot: the **determinant**. It measures how much the transformation *scales area*.



$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ stretches $\times 2$ across and $\times 3$ up: area $\times 6$.



A *shear* $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ slides the top over but keeps the area: still 1.

Computing the determinant of a 2×2

Definition

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is

$$\det A = ad - bc.$$

Its size $|\det A|$ is the area-scaling factor.

Example

$\det\begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix} = 3 \cdot 4 - 1 \cdot 2 = 12 - 2 = 10$. This matrix multiplies areas by 10.

When the determinant is zero, the matrix collapses space

Intuition

If $\det A = 0$, the area-scaling factor is 0: the transformation squashes the whole plane onto a single *line* (or a point). Different inputs get mapped to the same output, so you can no longer tell them apart — the matrix cannot be undone.



$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$: the two columns lie on one line, so the square is crushed onto it. $\det = 0$.

A zero determinant is the warning sign that a system may have *no* solution or *infinitely many*. A nonzero determinant guarantees exactly one.

Undoing a transformation: the inverse

Definition

The **inverse** A^{-1} undoes A : $A^{-1}A = I$. It exists exactly when $\det A \neq 0$ (the matrix didn't collapse anything). For a 2×2 matrix,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

and then $Ax = b$ has the unique solution $x = A^{-1}b$.

Example

Invert $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

First the determinant: $\det A = 1 \cdot 4 - 2 \cdot 3 = -2$. It is nonzero, so the inverse exists:

$$A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}.$$

You can check that $AA^{-1} = I$.

Solving larger systems by elimination

Gaussian elimination: use one equation to cancel a variable from the others, repeat, then back-substitute. It is the 2×2 idea, scaled up.

Example

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2.$$

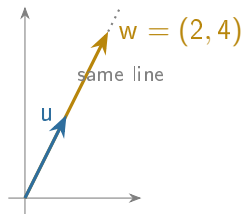
Removing x from the last two equations gives $3y + z = 9$ and $y - 2z = -4$. Substituting $z = 9 - 3y$ into the second: $7y = 14$, so $y = 2$, then $z = 3$ and $x = 1$. The solution is $(x, y, z) = (1, 2, 3)$.

When is a vector “redundant”?

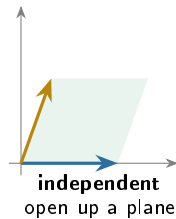
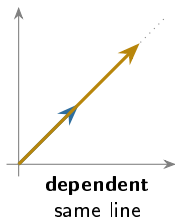
Look at $u = (1, 2)$ and $w = (2, 4)$.

w is just $2 \times u$ — it points in the *same* direction. Once you have u , adding w tells you nothing new: both lie on one line.

We say u and w are **linearly dependent**: one is a combination of the other.



Independent vs. dependent — the picture



Key idea

Vectors are **independent** when none is a combination of the others — each points in a genuinely new direction.

Span: what can you reach?

Definition

The **span** of some vectors is everything you can reach by linear combinations of them (scaling and adding).

- Two vectors on the *same line* span only that *line*.
- Two vectors in *different directions* span the whole *plane* — you can reach any point by mixing them.

So “how many independent directions you have” decides “how big a space you can reach”.

Rank = the number of independent directions

Key idea

The **rank** of a matrix is how many of its columns point in genuinely different directions — the number of independent directions in the data. It is the *honest* count of dimensions, which can be far fewer than the number of columns.

Example

$\begin{pmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \end{pmatrix}$: column 2 is $3\times$ column 1 (no new direction), but column 3 *is* a new direction. Two independent directions, so **rank** = 2.

Quick tests for independence

Useful fact

- Two vectors are dependent exactly when one is a multiple of the other.
- For a square matrix, the columns are independent exactly when $\det \neq 0$.
- Any more than n vectors in $\mathbb{R}^n$ are automatically dependent (you cannot have three independent directions in a plane).

Why rank matters in neuroscience

Link to cognitive science

This is exactly what we mean by the *dimensionality* of neural data. A recording of 100 neurons lives in $\mathbb{R}^{100}$, but their activity during a task often varies along only a handful of independent directions — the data have low rank.

Finding those few directions is *dimensionality reduction*, and the principled way to do it is Principal Component Analysis — exactly where next session begins.

Practice — rank and independence

Practice — try it

1. Are $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ independent?
2. Are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ independent?
3. Can three vectors in $\mathbb{R}^2$ ever be independent?

Session 3 — what we built

- Vectors are points/arrows/features, combined by *linear combination*.
- The *dot product* gives length, distance, angle, and similarity; *orthogonality* is “unrelated”; *projection* is least squares.
- A *matrix* is both a data table and a transformation; the *determinant* is its area-scaling factor.
- The *inverse* exists when $\det \neq 0$; *rank* counts the independent directions — the true dimensionality of data.

Next: the special directions of a matrix — eigenvectors — and their flagship application, PCA.